

The primary chemical composition of kidney stones commonly includes $\text{Ca}_3(\text{PO}_4)_2$ (calcium phosphate), CaC_2O_4 (calcium oxalate), and MgNH_4PO_4 (magnesium ammonium phosphate). The metal ions in these compounds are:

- ☐ A. alkali metals. [11%]
- ☐ B. strong Lewis acids. [14%]
- ☐ C. naturally occurring products of a two-electron reduction of elemental metals. [16%]
- ✓ ☒ D. formed by the two-electron oxidation of neutral metal atoms. [58%]

Correct

57% answered correctly

Explanation:

Typical charges of elements in different columns of the periodic table

Change needed for full valence shell	lose 1	lose 2	lose 3	lose 4	gain or lose 3	gain 3	gain 2	gain 1	gain none
Ionic charge result	1+	2+	3+	4+	3-	2-	1-	0	
	1A	2A	3A	4A	5A	6A	7A	8A	
	H•	Li• Be•	•B•	•C•	•N•	•O•	•F•	He•	
	Na• Mg•	Transition metals may have various charges		•Al•	•Si•	•P•	•S•	•Cl•	•Ar•
	K•	Ca•							

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Ionic compounds contain one or more ionic bonds, which result from **electrostatic attractions** between ions with opposite charges. The charges may be from individual ionized atoms or from **polyatomic ions**. An **ionic bond** between two atoms often involves the transfer of **valence electrons** from one atom (typically a metal) to another (typically a nonmetal) to create a cation and an anion.

Because of their relatively low **electronegativity** and **ionization energy**, metals tend to lose electrons to form cations whereas nonmetals tend to gain electrons to form anions. The charge of an atomic ion relates directly to the number of electrons gained or lost. For each **electron lost**, the **charge increases** by one unit, but for each **electron gained**, the **charge decreases** by one unit.

The electron-deficient character of metal cations causes them to be attracted to electron-rich ions or molecules and act as **Lewis acids**. The strength of the attraction increases as the ionic charge increases or as the **ionic radius** decreases. As a result, metal cations with a **smaller ionic radius** and a **higher positive charge** such as Al^{3+} and Ti^{4+} are **stronger Lewis acids** than metal cations with a larger ionic radius and lower charge such as Ca^{2+} and Mg^{2+} .

The alkaline-earth metals Ca and Mg each have two valence electrons, which are easily lost during oxidation in natural systems to form cations with a +2 charge.

(Choice A) Calcium and magnesium are alkaline-earth metals (Group 2), not **alkali metals** (Group 1).

(Choice B) The relatively large cationic radii and low charge of Ca^{2+} and Mg^{2+} make these ions **weak Lewis acids**.

(Choice C) Ca^{2+} and Mg^{2+} cations are produced by the *oxidation* of elemental Mg and Ca metals, not by **reduction**.

Educational objective:

Metals tend to lose electrons to form cations whereas nonmetals tend to gain electrons to form anions. The charge of an atom increases by one unit for each electron lost but decreases by one unit for each electron gained. Metal cations with a smaller ionic radius and a higher positive charge are stronger Lewis acids than those with a larger ionic radius and lower charge.

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General Chemistry

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Interactions of Chemical Substances